Pruning free resolutions over a complete intersection

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Setup

- $R = \mathbb{k}[x_1, \ldots, x_n]$ and $I \subseteq R$ a monomial ideal.
- $a_1, \ldots, a_c \in I$ a regular sequence, and $\overline{R} = R/(a_1, \ldots, a_c)$.
- Since $(a_1, \ldots, a_c) \subseteq I$, the third isomorphism theorem gives $R/I \cong \overline{R}/\big(I/(a_1, \ldots, a_c)\big)$, so R/I is an $\overline{R}$-module.

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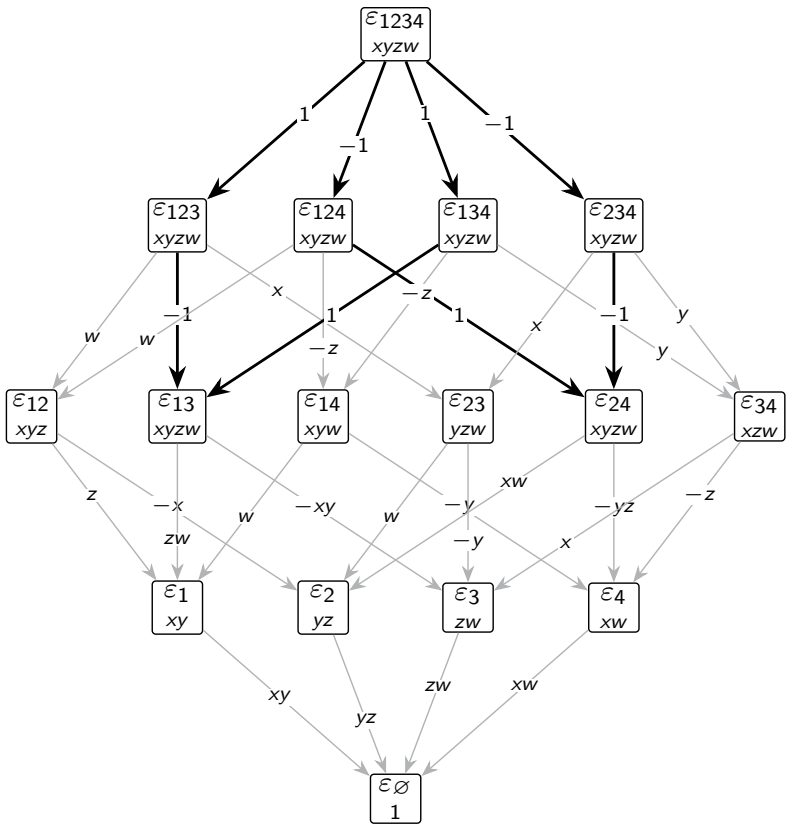
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- **Algebraic Morse theory** [Skö06, JW09]: find a matching on a graph that prunes a free resolution toward the minimal one.
- **Main theorem**: when $F_\bullet$ has a DG algebra structure, the two constructions commute.

Running example: the Taylor resolution

- $R = \mathbb{k}[x, y, z, w]$, $I = (m_1, \ldots, m_4) = (xy, yz, zw, xw)$. *Taylor resolution* $F_\bullet$: basis ε_S, $S \subseteq \{1, \ldots, 4\}$, in degree $|S|$, $m_S = \operatorname{lcm}(m_s : s \in S)$, $d(\varepsilon_S) = \sum_{s \in S} (-1)^{\#\{t \in S \,:\, t > s\}} \frac{m_S}{m_{S \setminus s}} \varepsilon_{S \setminus s}$.
- *Complex graph* $G(F_\bullet)$: an edge $\alpha \to \beta$ of weight $d_{\alpha,\beta}$ whenever it is nonzero.



$$d_4 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$$

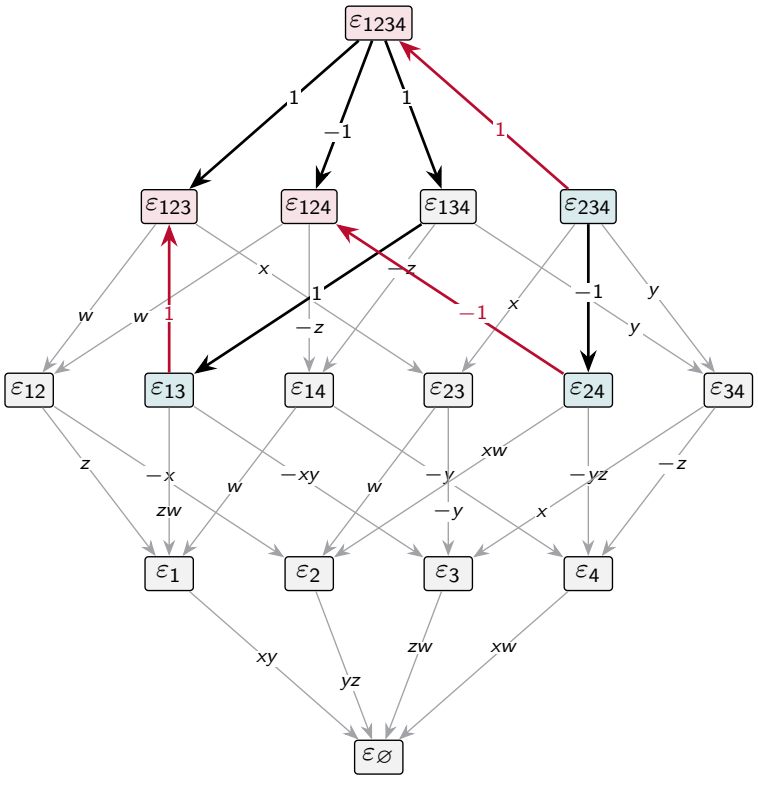
$$d_3 = \begin{pmatrix} w & w & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -z & -z & 0 \\ x & 0 & 0 & x \\ 0 & 1 & 0 & -1 \\ 0 & 0 & y & y \end{pmatrix}$$

$$d_2 = \begin{pmatrix} z & zw & w & 0 & 0 & 0 \\ -x & 0 & 0 & w & xw & 0 \\ 0 & -xy & 0 & -y & 0 & x \\ 0 & 0 & -y & 0 & -yz & -z \end{pmatrix}$$

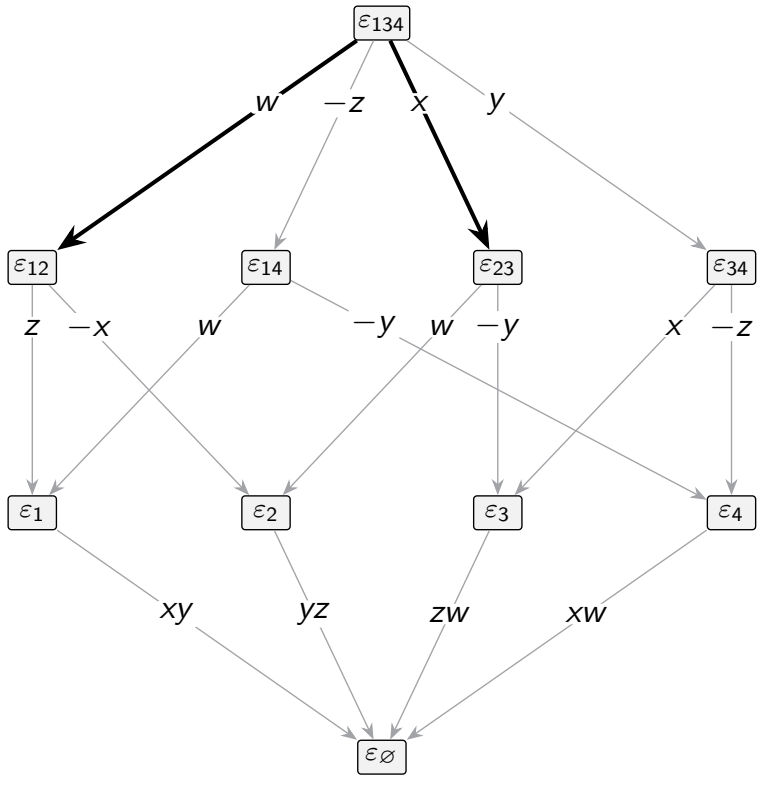
$$d_1 = \begin{pmatrix} xy & yz & zw & xw \end{pmatrix}$$

Running example: pruning

$G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$



$F_\bullet^{\mathcal{M}}$



- $\mathcal{M} = \{(\varepsilon_{123}, \varepsilon_{13}, -1), (\varepsilon_{124}, \varepsilon_{24}, 1), (\varepsilon_{1234}, \varepsilon_{234}, -1)\}$, edges $(\alpha, \beta, d_{\alpha,\beta})$, reversed in red with weights $-(d_{\alpha,\beta})^{-1}$.
- Bold: $\varepsilon_{134} \xrightarrow{1} \varepsilon_{13} \xrightarrow{1} \varepsilon_{123} \xrightarrow{w} \varepsilon_{12}$ and $\varepsilon_{134} \xrightarrow{1} \varepsilon_{13} \xrightarrow{1} \varepsilon_{123} \xrightarrow{x} \varepsilon_{23}$.

Algebraic Morse theory

- A *Morse matching* $\mathcal{M}$ on $G(F_\bullet)$: edges of unit weight, no two sharing a vertex, such that reversing them leaves no directed cycle. Unmatched vertices are *critical*, and span $F_\bullet{}^{\mathcal{M}}$.
- $d = \mu + \delta$: the *matched part* μ keeps the entries $d_{\alpha,\beta}$ with $(\alpha, \beta) \in \mathcal{M}$, and $\delta = d - \mu$.
- A *gradient path* alternates an edge of δ, going down, with a reversed matched edge $\beta \to \alpha$ of weight $-(d_{\alpha,\beta})^{-1}$, going up. Its weight is the product of its edge weights.

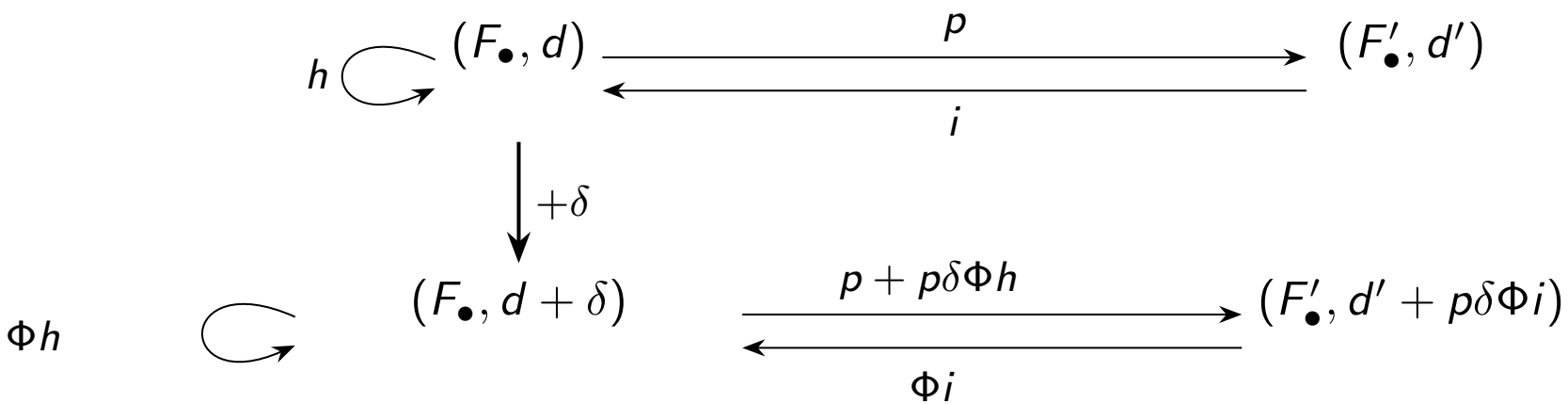
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- Homological degree 2, $h(\beta) = -(d_{\alpha,\beta})^{-1}\alpha$:

$$i|_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad p|_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad h|_2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

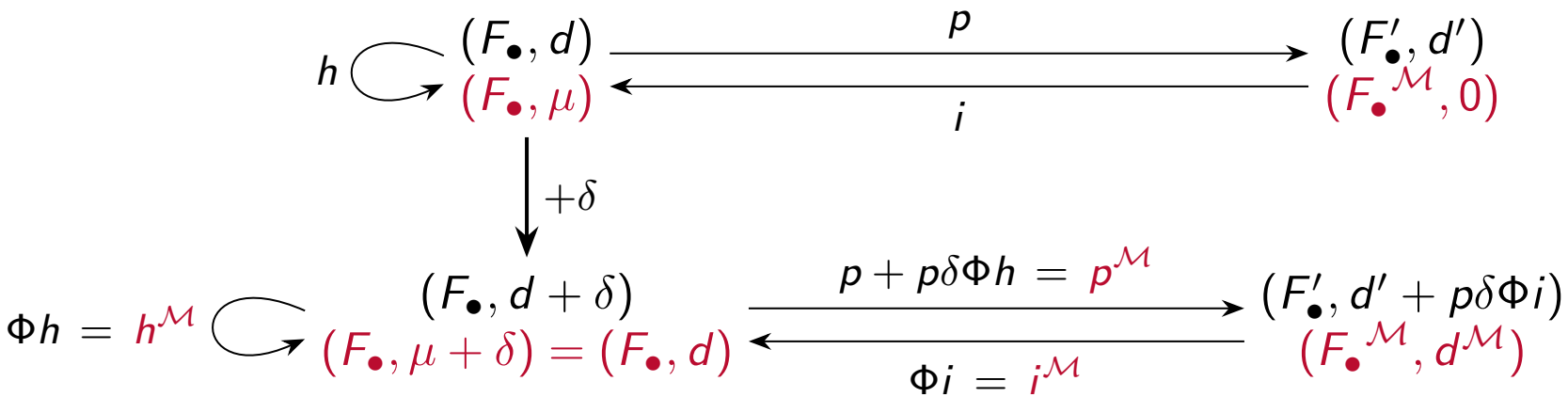
The perturbation lemma and AMT

- A *contraction* (i, p, h) of $(F_\bullet, d)$ onto $(F'_\bullet, d')$: chain maps i, p and a linear map h of degree $+1$ with $pi = \mathrm{id}$, $ip - \mathrm{id} = dh + hd$, $h^2 = hi = ph = 0$.
- **Lemma.** If $(d + \delta)^2 = 0$ and $\Phi = \sum_{n \geq 0} (h\delta)^n = (\mathrm{id} - h\delta)^{-1}$ is finite on each element, the bottom row is a contraction.



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- **Theorem** [Skö06, JW09]. $(F_\bullet{}^{\mathcal{M}}, d^{\mathcal{M}})$ is a free resolution, with $d^{\mathcal{M}} = p\delta\Phi i = p(\delta + \delta h\delta + \delta h\delta h\delta + \cdots)i$ summing over gradient paths.

Systems of higher homotopies

- $F_\bullet$ a free resolution of R/I over R with differential d, and $a_1, \dots, a_c \in I$ a regular sequence, $\overline{R} = R/(a_1, \dots, a_c)$.
- A *system of higher homotopies* for $a_1, \dots, a_c$ on $F_\bullet$ [Eis80]: maps $\sigma_{\boldsymbol{u}} \colon F_\bullet \to F_{\bullet+2|\boldsymbol{u}|-1}$, $\boldsymbol{u} \in \mathbb{N}^c$, with $\sigma_{\mathbf{0}} = d$ and

$$\sum_{\boldsymbol{v}+\boldsymbol{v}'=\boldsymbol{u}} \sigma_{\boldsymbol{v}}\sigma_{\boldsymbol{v}'} = \begin{cases} a_j \,\mathrm{id} & \boldsymbol{u} = \boldsymbol{e}_j, \\ 0 & \text{otherwise.} \end{cases}$$

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- Truncated: $\sigma_{\boldsymbol{u}} = 0$ for $|\boldsymbol{u}| \geq 2$. With $\sigma^{(j)} = \sigma_{\boldsymbol{e}_j}$ this says

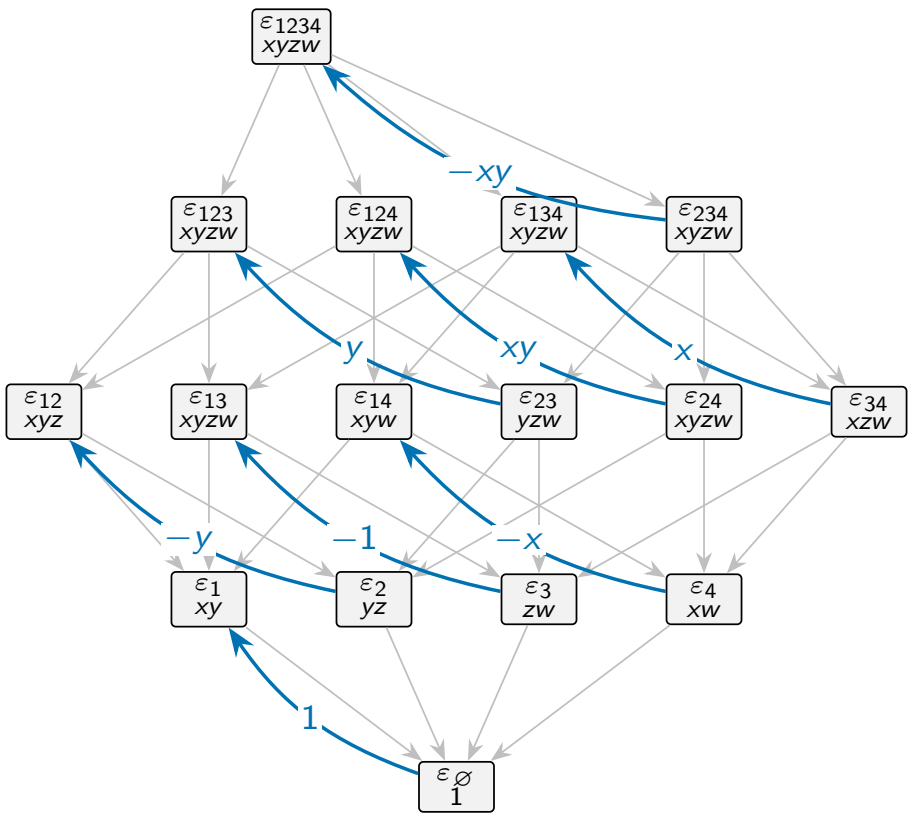
$$d\sigma^{(j)} + \sigma^{(j)}d = a_j \operatorname{id}, \qquad (\sigma^{(j)})^2 = 0, \qquad \sigma^{(j)}\sigma^{(k)} + \sigma^{(k)}\sigma^{(j)} = 0.$$

- **DG resolutions**: if $F_\bullet$ is a strictly-graded commutative DG algebra, choose $z_j \in F_1$ with $dz_j = a_j$. Multiplication by the z_j is a truncated system of higher homotopies

Sobieska's homotopies on the Taylor resolution

- [Sob23] For $F_\bullet$ the Taylor resolution of $I = (m_1, \ldots, m_r)$ and $a_j = \sum_t f_{j,t} m_t$, set $\sigma_{\boldsymbol{u}} = 0$ for $|\boldsymbol{u}| \geq 2$ and

$$\sigma^{(j)}(\varepsilon_S) = \sum_{t \notin S} (-1)^{\#\{s \in S \,:\, s > t\}} f_{j,t} \gcd(m_t, m_S)\, \varepsilon_{S \cup t}.$$



Running example, $a = xy$. Blue: σ.

The Eisenbud–Shamash resolution

- $R\{y_i\}_{i=1}^c$: the graded dual of $R[t_1, \dots, t_c]$, with each t_j in homological degree -2. It has R-basis $y^{\boldsymbol{u}} = y_1^{(u_1)} \cdots y_c^{(u_c)}$, $\boldsymbol{u} \in \mathbb{N}^c$, dual to the monomials $t^{\boldsymbol{u}}$, with $y^{\boldsymbol{u}}$ in degree $2|\boldsymbol{u}|$. The *CI operator* t_j sends $y^{\boldsymbol{u}}$ to $y^{\boldsymbol{u}-\boldsymbol{e}_j}$, or to 0 if $u_j = 0$.
- Over R, on $R\{y_i\}_{i=1}^c \otimes_R F_\bullet$:

$$D = \sum_{\boldsymbol{u}} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}}, \qquad D^2 = \sum_j t_j \otimes a_j \,\mathrm{id}\,.$$

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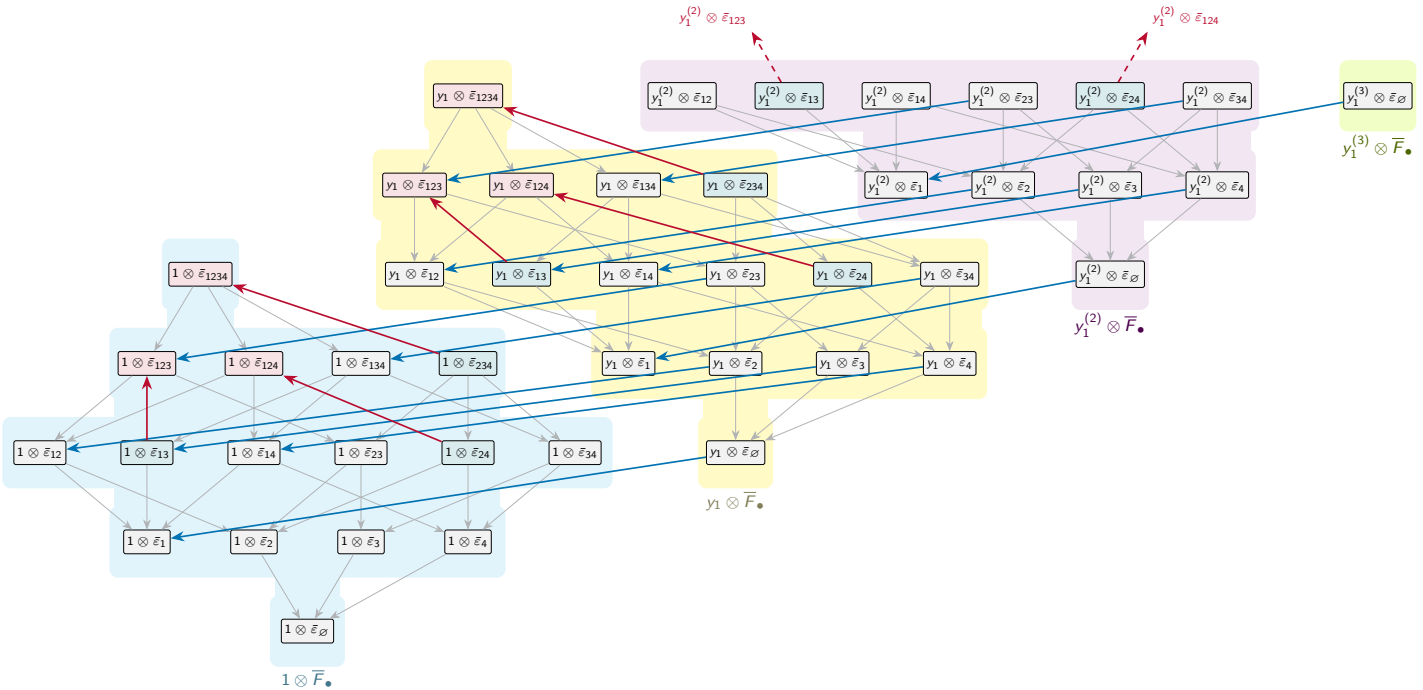
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- A bar is reduction modulo $(a_1, \dots, a_c)$: $\overline{F}_\bullet = F_\bullet \otimes_R \overline{R}$ and $\bar{f} = f \otimes \mathrm{id}$.
- **Theorem** [Eis80]. $\mathrm{Sh}(F_\bullet, \sigma) = \left(R\{y_i\}_{i=1}^c \otimes_R \overline{F}_\bullet, \overline{D}\right)$ is a free $\overline{R}$-resolution of R/I.
- Truncated: $D = \partial_0 + \partial_1 + \cdots + \partial_c$ with $\partial_0 = \mathrm{id} \otimes d$ and $\partial_j = t_j \otimes \sigma^{(j)}$.

Lifting the matching

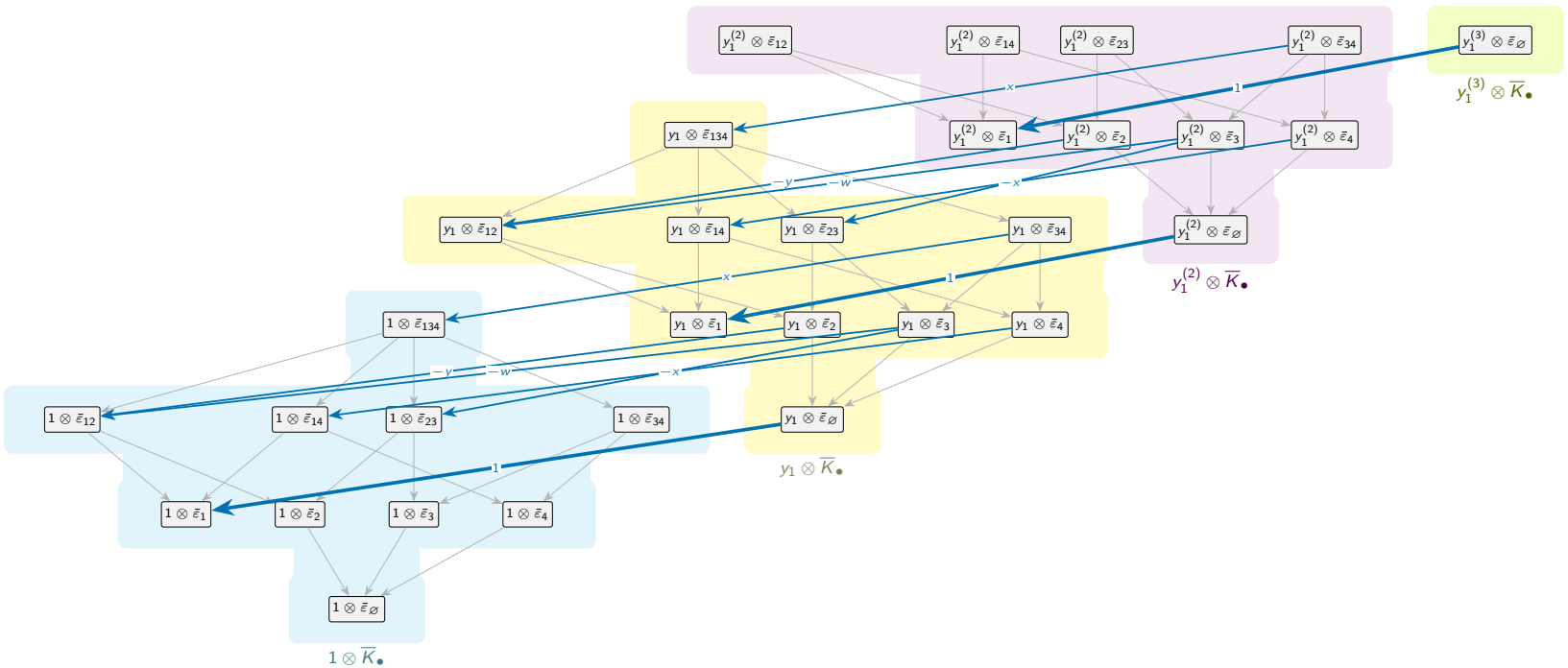
- $c = 1$, $a = xy$, $\overline{R} = R/(xy)$: copies $y_1^{(k)} \otimes \overline{F}_\bullet$, $k \geq 0$. Blue: $\sigma = \varepsilon_1 \cdot (-)$.
- $\overline{\mathcal{M}} = \{(y_1^{(k)} \otimes \bar{\varepsilon}_{123}, y_1^{(k)} \otimes \bar{\varepsilon}_{13}),\ (y_1^{(k)} \otimes \bar{\varepsilon}_{124}, y_1^{(k)} \otimes \bar{\varepsilon}_{24}),$
  $(y_1^{(k)} \otimes \bar{\varepsilon}_{1234}, y_1^{(k)} \otimes \bar{\varepsilon}_{234}) : k \geq 0\}$.



- Ranks $1, 4, 7, 8, 8, \ldots$

Result of pruning

- $c = 1$, $a = xy$, $\overline{R} = R/(xy)$: copies $y_1^{(k)} \otimes \overline{F}_\bullet$, $k \geq 0$. Blue: $\sigma = \varepsilon_1 \cdot (-)$.
- The Morse complex $\mathrm{Sh}(F_\bullet, \sigma)^{\overline{\mathcal{M}}}$.



- Ranks $1, 4, 5, 5, 5, \ldots$
- Still not minimal :(

Transferred system of higher homotopies

- The perturbed contraction of algebraic Morse theory, with $\Phi = \sum_{n\geq 0}(h\delta)^n$:

$$(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}}) = \big(\Phi i,\ p + p\delta\Phi h,\ \Phi h\big), \qquad h^{\mathcal{M}} = \sum_{n\geq 0}(h\delta)^n h.$$

- Set $\sigma_0^{\mathcal{M}} = d^{\mathcal{M}}$ and, for $|\boldsymbol{u}| = k \geq 1$,

$$\sigma_{\boldsymbol{u}}^{\mathcal{M}} = \sum_{\boldsymbol{e}_{j_1}+\cdots+\boldsymbol{e}_{j_k}=\boldsymbol{u}} p^{\mathcal{M}}\sigma^{(j_1)}h^{\mathcal{M}}\sigma^{(j_2)}\cdots h^{\mathcal{M}}\sigma^{(j_k)}i^{\mathcal{M}}.$$

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- New blue arrows:

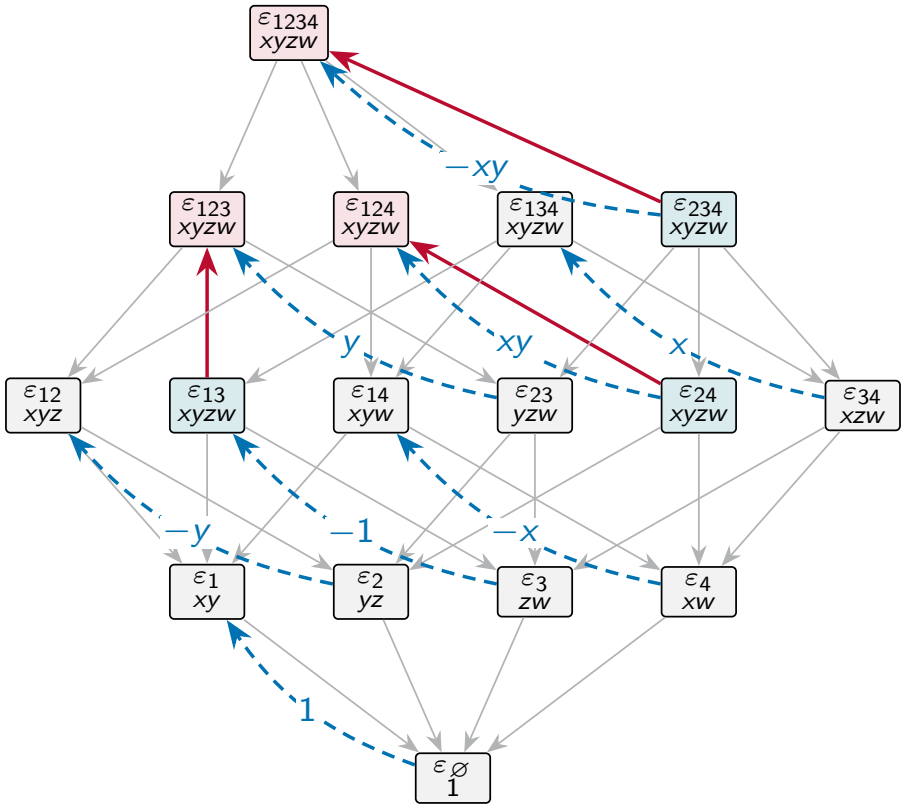
$$\begin{aligned}\sigma^{\mathcal{M}}(\varepsilon_3) &= p^{\mathcal{M}}\sigma i^{\mathcal{M}}(\varepsilon_3) = p^{\mathcal{M}}\sigma(\varepsilon_3) = p^{\mathcal{M}}(-\varepsilon_{13})\\ &= -p\delta h(\varepsilon_{13}) = -p\delta(\varepsilon_{123}) = -w\varepsilon_{12} - x\varepsilon_{23},\end{aligned}$$

from the gradient paths $\varepsilon_{13} \xrightarrow{1} \varepsilon_{123} \xrightarrow{w} \varepsilon_{12}$ and $\varepsilon_{13} \xrightarrow{1} \varepsilon_{123} \xrightarrow{x} \varepsilon_{23}$.

Main theorem

Theorem (B.–Vassilev [BV26]). Let σ be a truncated system of higher homotopies on $F_\bullet$ and $\mathcal{M}$ a Morse matching on $F_\bullet$. Then $\sigma^{\mathcal{M}}$ is a system of higher homotopies on $F_\bullet{}^{\mathcal{M}}$, and

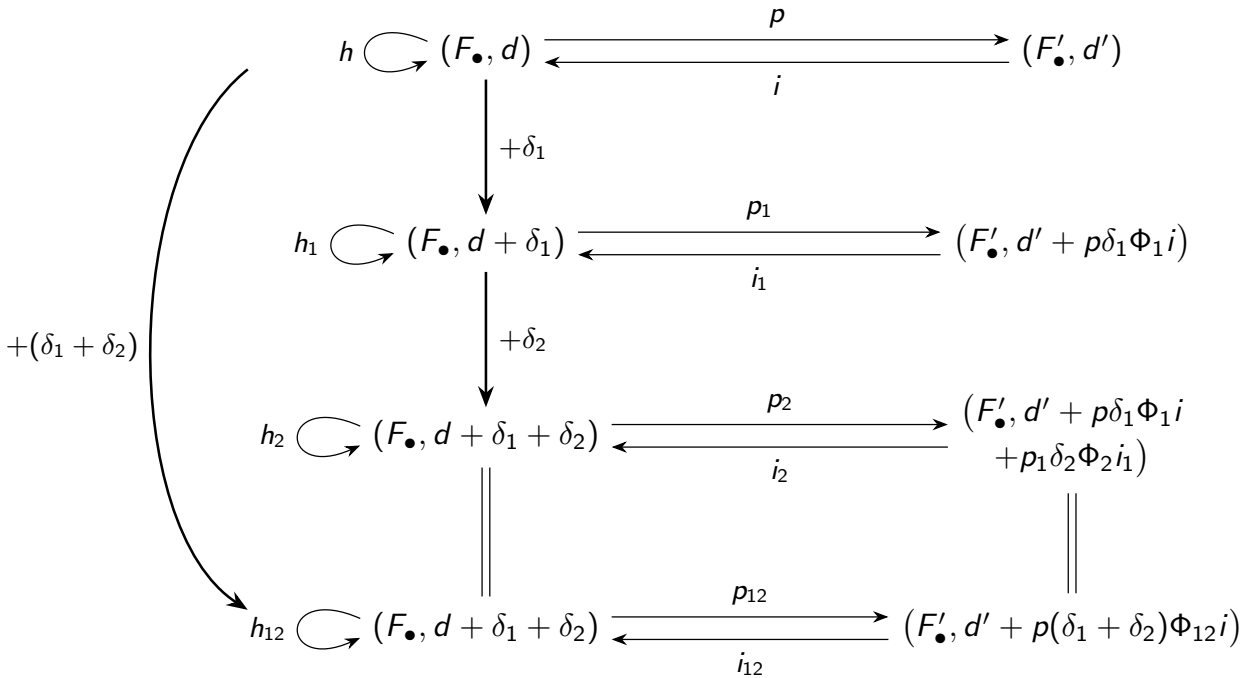
$$\mathrm{Sh}(F_\bullet, \sigma)^{\overline{\mathcal{M}}} = \mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}}).$$



$$\sigma^{\mathcal{M}}_{\boldsymbol{u}} = \sum p^{\mathcal{M}} \sigma^{(j_1)} h^{\mathcal{M}} \cdots h^{\mathcal{M}} \sigma^{(j_k)} i^{\mathcal{M}}$$

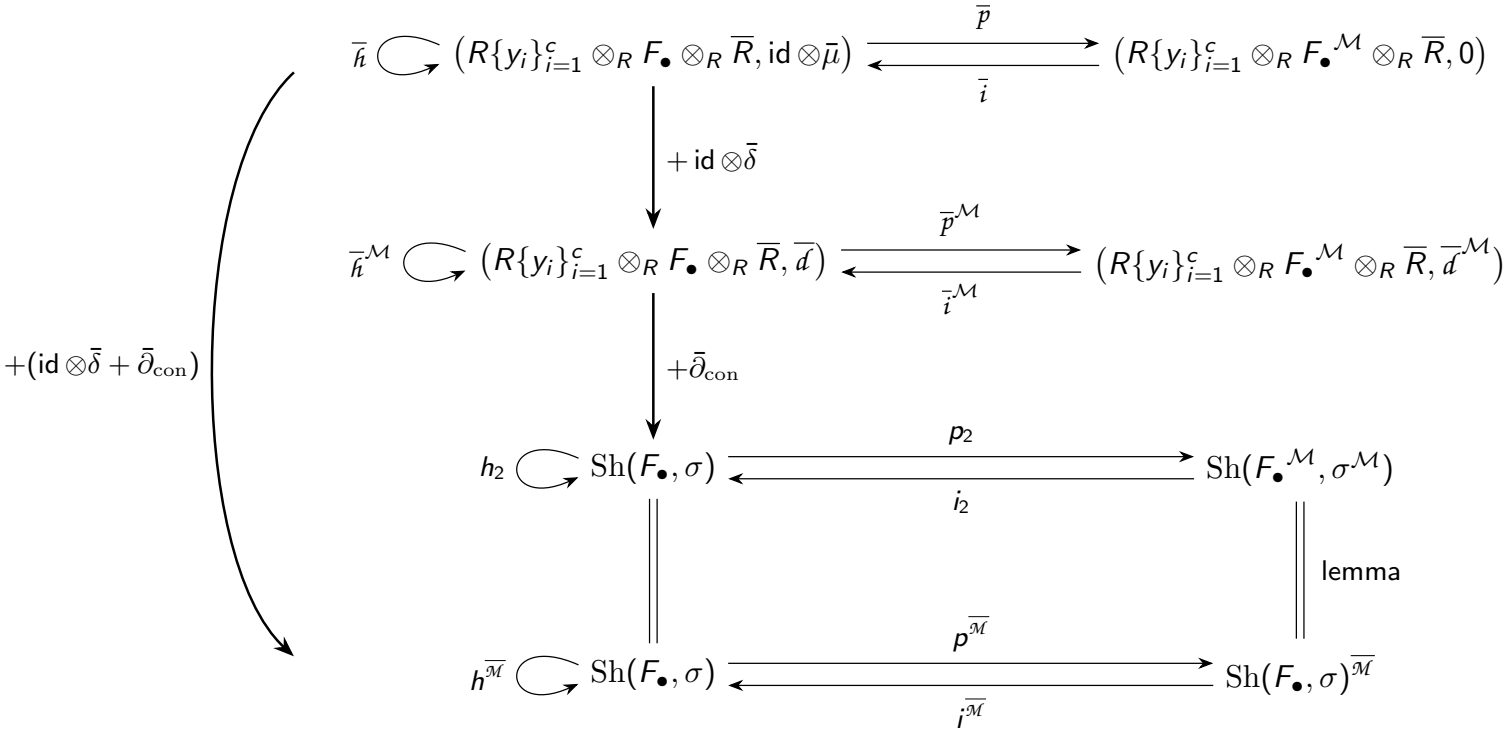
σ: up a blue edge. $h^{\mathcal{M}} = \Phi h$: up a red edge, then along gradient paths.

Main tool for the proof



- (i, p, h) a contraction of $(F_\bullet, d)$ onto $(F'_\bullet, d')$, and δ_1, $\delta_1 + \delta_2$ perturbations of d, with these series finite on each element:
  - $\Phi_1 = \sum_{n\geq 0}(h\delta_1)^n$, which gives the contraction (i_1, p_1, h_1) after perturbing by δ_1,
  - $\Phi_2 = \sum_{n\geq 0}(h_1\delta_2)^n$,
  - $\Phi_{12} = \sum_{n\geq 0}(h(\delta_1+\delta_2))^n$.
- **Lemma**. Perturbing by δ_1 and then by δ_2 gives the same differential as perturbing by $\delta_1 + \delta_2$.

Proof



- Script letters are lifts: $\mathcal{f} = \mathrm{id}\otimes f$, one copy of f over each $\boldsymbol{u}$.

Future work

- **Question.** Does the main theorem extend to systems of higher homotopies that are not truncated: is $\mathrm{Sh}(F_\bullet, \sigma)^{\overline{\mathcal{M}}}$ again an Eisenbud–Shamash resolution $\mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})$?
  $\partial_{\mathrm{con}} = \sum_{\boldsymbol{u}\neq\mathbf{0}} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}}$ jumps several levels: no splitting $\partial_1 + \cdots + \partial_c$.
- **Question.** Can the minimal free resolution of R/I over $\overline{R}$ be reached from $\mathrm{Sh}(F_\bullet, \sigma)$ by finitely many Morse matchings?
- **Question.** If $F_\bullet$ is cellular, supported on a CW complex, can the gluing of the copies $y^{\boldsymbol{u}} \otimes \overline{F}_\bullet$ be made to respect that cell structure?

References

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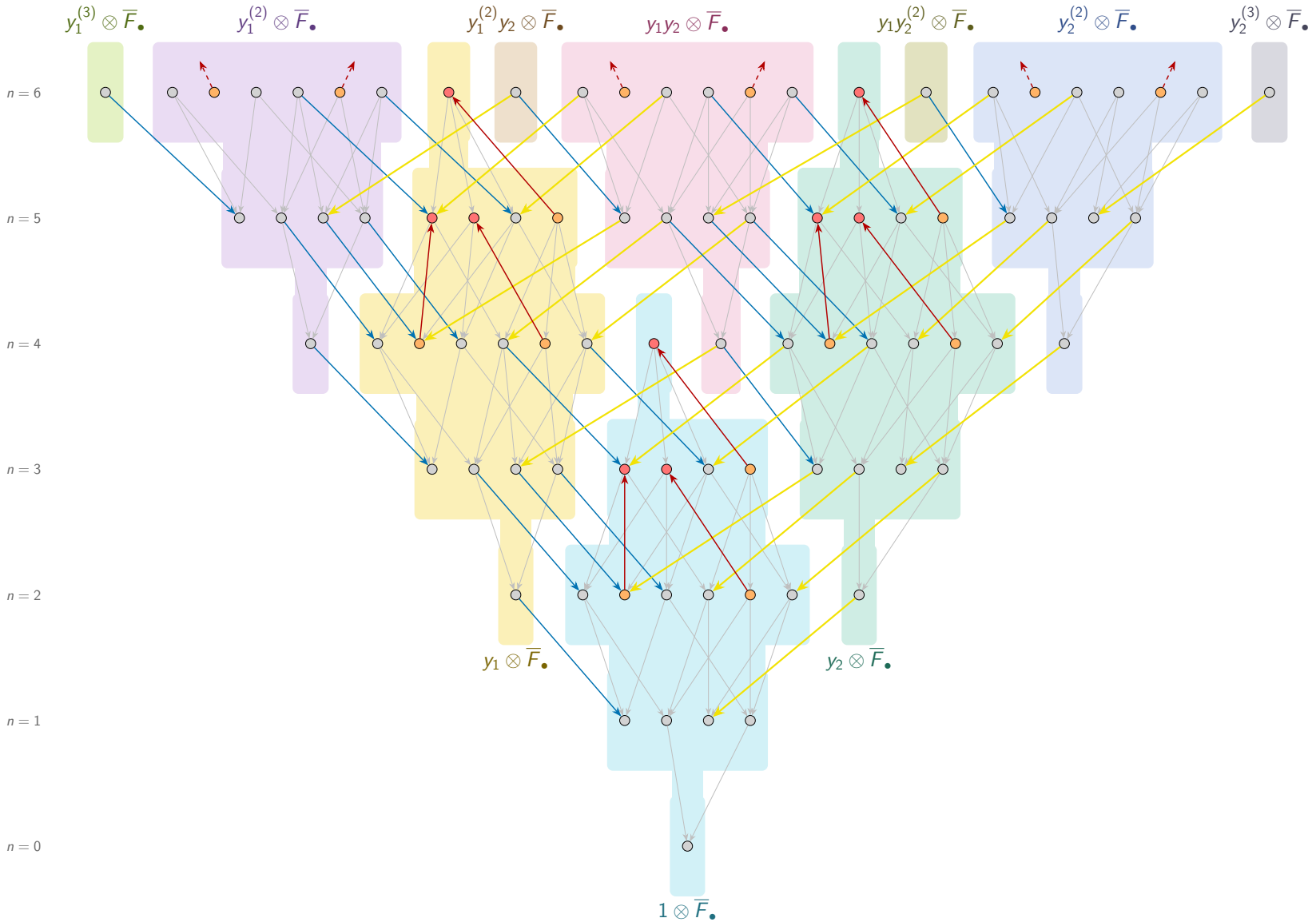
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[Sob23] Aleksandra Sobieska, *A Taylor resolution over complete intersections*, J. Algebra **636** (2023), 716–731.

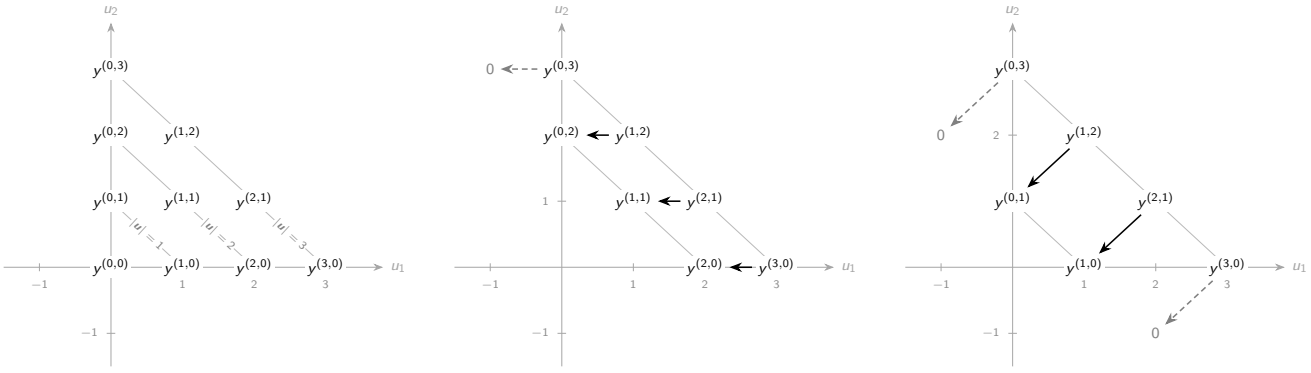
Thank you!

The case $c = 2$



$$R = \mathbb{k}[x, y, z, w],\ I = (xy, yz, zw, xw),\ a_1 = xy,\ a_2 = zw.$$

Shamash differential



(a) $t^{(0,0)}$ (b) $t^{(1,0)}$ (c) $t^{(1,1)}$

$$T^{(3)}_{(0,0)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad T^{(3)}_{(1,0)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \qquad T^{(3)}_{(1,1)} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$D|_6 =$

	$k=0$	$k=1$	$k=2$	$k=3$
$k'=0$	d	$\left[\begin{array}{c\|c} \sigma_{(1,0)} & \sigma_{(0,1)} \end{array}\right]$	$\left[\begin{array}{c\|c\|c} \sigma_{(2,0)} & \sigma_{(1,1)} & \sigma_{(0,2)} \end{array}\right]$	$\left[\begin{array}{c\|c\|c\|c} \sigma_{(3,0)} & \sigma_{(2,1)} & \sigma_{(1,2)} & \sigma_{(0,3)} \end{array}\right]$
$k'=1$	0	$\left[\begin{array}{c\|c} d & 0 \\ \hline 0 & d \end{array}\right]$	$\left[\begin{array}{c\|c\|c} \sigma_{(1,0)} & \sigma_{(0,1)} & 0 \\ \hline 0 & \sigma_{(1,0)} & \sigma_{(0,1)} \end{array}\right]$	$\left[\begin{array}{c\|c\|c\|c} \sigma_{(2,0)} & \sigma_{(1,1)} & \sigma_{(0,2)} & 0 \\ \hline 0 & \sigma_{(2,0)} & \sigma_{(1,1)} & \sigma_{(0,2)} \end{array}\right]$
$k'=2$	0	0	$\left[\begin{array}{c\|c\|c} d & 0 & 0 \\ \hline 0 & d & 0 \\ \hline 0 & 0 & d \end{array}\right]$	$\left[\begin{array}{c\|c\|c\|c} \sigma_{(1,0)} & \sigma_{(0,1)} & 0 & 0 \\ \hline 0 & \sigma_{(1,0)} & \sigma_{(0,1)} & 0 \\ \hline 0 & 0 & \sigma_{(1,0)} & \sigma_{(0,1)} \end{array}\right]$

- $\left(D|_n\right)_{k,k'} = \sum_{|\boldsymbol{v}|=k-k'} T^{(k)}_{\boldsymbol{v}} \otimes \sigma_{\boldsymbol{v}}|_{n-2k}$ for $k' \leq k$, and 0 for $k' > k$.